

Question 1

Question ID: 1575129

Which derivative markets feature novation by a central clearinghouse?

A) All exchange-traded and over-the-counter derivatives markets.



B) Exchange-traded derivatives markets, but not over-the-counter derivatives markets.



C) All exchange-traded derivatives markets and some over-the-counter derivatives markets.



Explanation

Novation refers to a central clearinghouse taking the opposite side of each trade. All exchange-traded derivatives markets feature clearinghouse novation, as do over-the-counter derivatives markets that are subject to a central clearing mandate. (Module 68.1, LOS 68.b)

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Question 2

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Cash flows related to futures margin *least likely* include:

A) interest on the margin loan.



B) deposits to meet margin calls.



C) interest received on collateral.



Explanation

Futures margin is satisfied by posting collateral and does not involve a loan. A futures investor may post interest-bearing securities as collateral and earn interest (collateral yield) on these securities. Faced with a margin call, a futures investor must either post additional margin to restore the account to the initial margin requirement or close the position. (Module 69.1, LOS 69.a)

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Question 3

Question ID: 1575131

Fred's Securities writes a call option on Acme stock with an exercise price of \$40. The call option premium is \$5. On the expiration date, the price of Acme stock is \$43. Fred's profit or loss on this position is a:

A) loss of \$2.



B) loss of \$5.



C) profit of \$2.



Explanation

As the writer of the option, Fred's collects the \$5 premium. At expiration, the call option is in the money by $\$43 - \$40 = \$3$ and the option holder will exercise. Fred's must deliver a share of Acme to the option holder and receive \$40 when the share is worth \$43. The net result for Fred's Securities is $\$5 + \$40 - \$43 =$ a profit of \$2. (Module 69.2, LOS 69.b)

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Question 4

Question ID: 1575132

Which of the following parties to an over-the-counter derivatives contract *least likely* faces counterparty risk, assuming the contract is not subject to a central clearing mandate?

A) The writer of a put option.



B) The seller in a forward contract.



C) The pay-floating party in a swap.



Explanation

The writer of an option receives a premium at initiation and is not owed any further cash from the option buyer, regardless of whether the buyer exercises the option. In a forward contract or a swap, both parties face counterparty risk unless the contract is subject to a central clearing mandate. (Module 70.1, LOS 70.a)

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Question 5

Question ID: 1575133

Which of the following derivatives positions replicates investing at the risk-free rate?

A) Holding an asset and a short position in a forward contract on the asset.



B) Holding an asset and a long position in a forward contract on the asset.



C) Selling an asset short and holding a short position in a forward contract on the asset.



Explanation

Holding an asset and a short position in a forward contract on the asset replicates investing at the risk-free rate because the future payoff is certain. (Module 71.1, LOS 71.a)

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Question 6

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Compared to an asset with no net cost of carry, holding costs that are greater than benefits:

A) increase the no-arbitrage price of the forward contract.



B) decrease the no-arbitrage price of the forward contract.



C) have no effect on the no-arbitrage price of the forward contract.



Explanation

Costs of holding the underlying asset that are greater than the benefits increase the no-arbitrage price of a forward contract. (Module 71.1, LOS 71.b)

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Question 7

Question ID: 1575135

An implied 1-year forward rate two years from now can be determined based on:

A) 1-year and 2-year spot rates.



B) 1-year and 3-year spot rates.



C) 2-year and 3-year spot rates.



Explanation

The implied 1-year rate two years from now is the rate that equates borrowing or lending at the 3-year spot rate with borrowing or lending at the 2-year spot rate and then for one year at the forward rate. (Module 72.1, LOS 72.b)

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Question 8

Question ID: 1575136

The price of an interest rate futures contract is *most likely* to differ from the price of an otherwise identical interest rate forward contract if:

A) their prices exhibit convexity bias.



B) interest rates are uncorrelated with futures prices.



C) the forward contract requires daily mark-to-market cash flows.



Explanation

Convexity bias can cause interest rate forward and futures prices to differ significantly for contracts with long maturities. Forward and futures prices may differ if a forward contract does not require daily cash settlement of mark-to-market gains and losses (as futures contracts require), and interest rates are positively or negatively correlated with futures prices over time. (Module 73.1, LOS 73.b)

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Question 9

Question ID: 1575137

In which of the following ways is an interest rate swap different from a series of forward rate agreements (FRAs)?

A) The FRAs that replicate an interest rate swap have nonzero values at initiation.



B) The fixed rate is known at initiation for an interest rate swap but not for a series of FRAs.



C) In a swap both parties pay cash each period, while in a series of FRAs the parties only exchange the net cash flows.



Explanation

An interest rate swap may be replicated by a series of FRAs with nonzero values at initiation, if their present values sum to zero at initiation. Typically the parties exchange net cash flows in either a swap or a series of FRAs. The fixed rate is known at initiation for either an interest rate swap or a series of FRAs. (Module 74.1, LOS 74.a)

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Question 10

Question ID: 1575138

The price of a swap contract:

A) remains constant throughout its life.



B) equals the present value of the expected future floating-rate payments minus the present value of the future fixed-rate payments.



C) equals the present value of the future fixed-rate payments minus the present value of the expected future floating-rate payments.



Explanation

The *price* of a swap contract is the par swap rate, which is the fixed rate that gives the swap a value of zero at initiation. The *value* of a swap contract to the fixed-rate payer is the present value of the expected future floating-rate payments minus the present value of the future fixed-rate payments. (Module 74.1, LOS 74.b)

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Question 11

Question ID: 1575139

Reasons why the no-arbitrage approach to pricing options is different from the no-arbitrage approach to pricing forward commitments *least likely* include that:

A) options have one-sided payoffs, while forward commitments have two-sided payoffs.



B) Many forward commitments are exchange-traded derivatives, while most options are over-the-counter derivatives.



C) options have positive values at initiation, while forward commitments typically have a value of zero at initiation.



Explanation

Arbitrage and replication concepts apply to both exchange-traded and over-the-counter derivatives. The differences between the models for forward commitments and options arise from the facts that initial values are positive for options and typically zero for forward commitments, and that gains and losses are one-sided for options and two-sided for forward commitments. (Module 75.1, LOS 75.b)

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Question 12

Question ID: 1575140

The value of a call option on a stock is *most likely* to decrease as a result of:

A) an increase in asset price volatility.



B) a decrease in the risk-free rate of interest.



C) a decrease in the exercise price of the option.



Explanation

A decrease in the risk-free rate of interest will decrease call values. The other changes will tend to increase the value of a call option. (Module 75.1, LOS 75.c)

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Question 13

Question ID: 1575141

A European call option is equivalent to a portfolio consisting of:

A) a short position in a forward contract, a long position in a put option, and a short position in the risk-free asset.



B) a long position in a forward contract, a short position in a put option, and a long position in the risk-free asset.



C) a long position in a forward contract, a long position in a put option, and a short position in the risk-free asset.



Explanation

The put-call forward parity relationship, at contract initiation, is $F_0(T)(1 + Rf)^{-T} + p_0 = c_0 + X(1 + Rf)^{-T}$. Therefore, the payoffs from a call option can be replicated with $c_0 = F_0(T)(1 + Rf)^{-T} + p_0 - X(1 + Rf)^{-T}$, or a long position in a forward contract, a long position in a put option, and a short position in the risk-free asset. (Module 76.1, LOS 76.b)

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